

- A cyclist rides along a straight road from A to a point B. He starts from rest at A and accelerates uniformly to reach a speed of 12 ms^{-1} in 8 seconds. He maintains this speed for a further 20 seconds and then uniformly decelerates to rest at B. If the whole journey takes 34 seconds, draw a velocity-time graph for the motion and from it find:
 - his acceleration for the first part of the motion,
 - his acceleration for the last part of the motion,
 - the total distance travelled.
- A battery operated toy dog starts at a point O and moves in a straight line. Its motion is modelled by the velocity-time graph in Fig. 2.

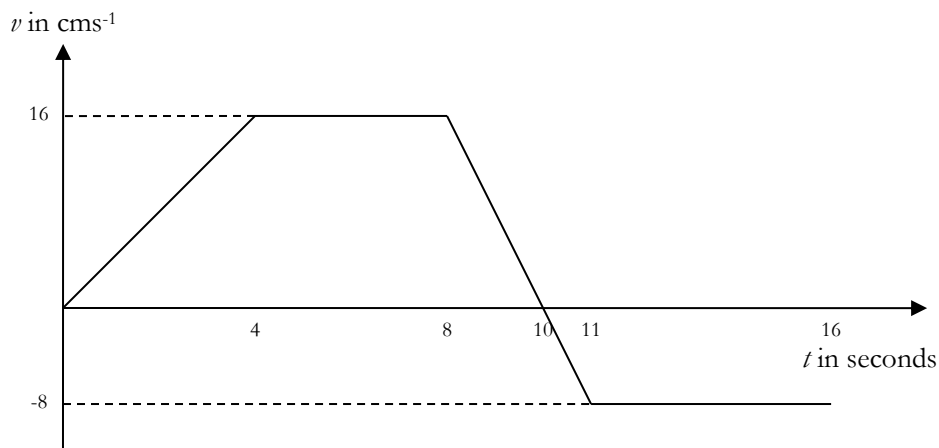


Fig. 2

- Calculate the displacement from O of the toy
 - after 10 seconds,
 - after 16 seconds.
 - Write down expressions for the velocity of the toy at time t seconds in the intervals $0 \leq t \leq 4$ and $4 \leq t \leq 8$.
 - Find the displacement from O of the toy after 1, 2, 3 and 3.5 seconds. In general, find the displacement from O at time t seconds in the interval $0 \leq t \leq 4$. Sketch the displacement time graph.
- A bullet is fired from a point O into a sand bank and moves in a straight line. Its motion is modelled by the velocity-time graph in Fig 3.

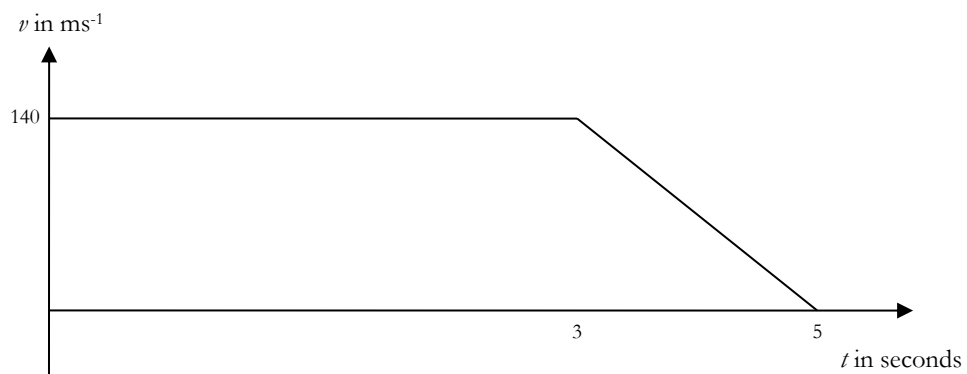


Fig. 3

- How far is the sand bank from O?
- How far into the sand bank does the bullet penetrate?
- Write down expressions for the velocity of the bullet at time t seconds in the intervals $0 \leq t \leq 3$ and $3 \leq t \leq 5$.

4. A bus is travelling at a constant speed of 10 ms^{-1} . A car starts moving just as the bus passes it and accelerates to 15 ms^{-1} in 20 seconds. 2 minutes later the car has brake to stop at a set of traffic lights whereas the bus can go straight through in the bus lane. Their motion is modelled by the velocity-time graph in Fig. 4.

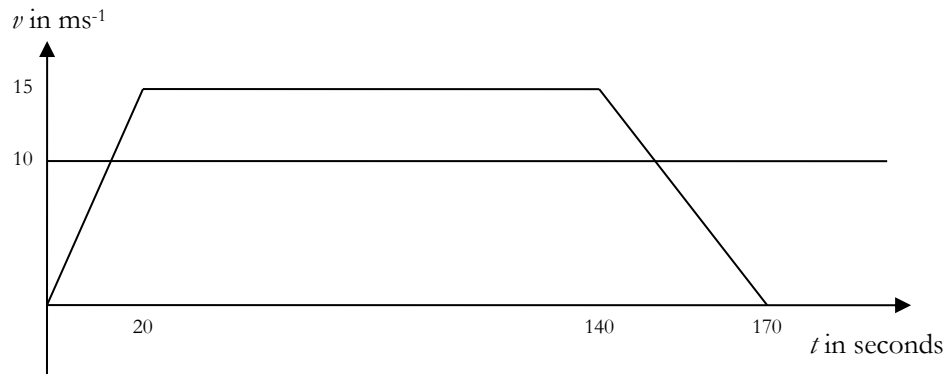
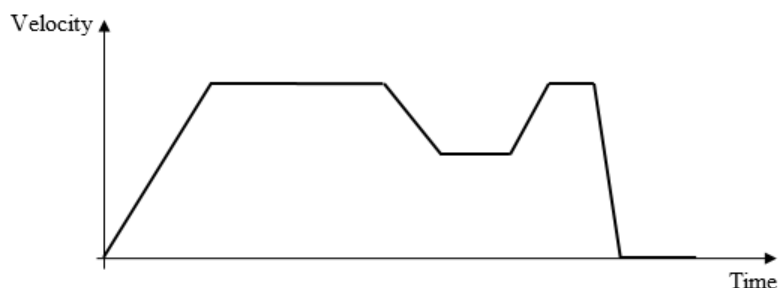


Fig. 4

- (a) How far are the traffic lights from the cars starting point?
 - (b) How far does the bus travel in the first 170 seconds?
 - (c) When does the bus pass the traffic lights?
 - (d) How far in front of the bus is the car at 140 s?
5. My flashy new sports car claims that it can go from 0-60 mph in 7.4 seconds.
- (a) Find the acceleration of my car in ms^{-2} , assuming that it is constant.
 - (b) How quickly would my car cover a $\frac{1}{8}$ of a mile drag racing strip, accelerating at this constant rate from rest?
 - (c) Why is my model of constant acceleration unrealistic?
6. A bus accelerates from stationary at 3 ms^{-2} for 6 seconds.
- (a) How far does it travel?
 - (b) In exactly this time, a parcel of mass 6 kg slides unimpeded the full length of the bus (10 m).
 - (c) Draw displacement-time and velocity-time graphs for this situation, assuming constant acceleration for each of the parcel and the bus.
 - (d) Calculate the frictional force acting upon the parcel by the bus.
7. Below is a graph showing velocity against time.



- a. Write a story that fits this velocity-time graph.
There is no scale so you can decide what the speeds are.
- b. Sketch a displacement-time graph for the journey.